

Video sharing platform

A Project Synopsis Report Submitted To



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1. Problem Statement.

Most consumer facing video sharing services today are full, algorithmically curated free-for-all marketing themselves relentlessly. As a result, students and local organizations in need of an affordable dedicated place to present their content are challenged; Rather than simply focusing on the distribution of content in siloed silos, these systems require users to traverse vast and complex networks filled with video-sharing. This strongly suggests the need for a simple, easy-to-use online application that minimises these unnecessary complexities by focusing on only what you really need.

2. Introduction

These are a types of web application that allows the user to upload videos and share or watch it online. In a fashion closer to the more famous sites such as YouTube, this project proposes an easy system in which users can sign up for accounts and upload their videos, listen/watch other people's uploaded videos. Users can also like, comment on, and share videos with other users via the platform.

3. Objective

- Develop a platform that allows users to easily upload and view videos.
- User account and profile feature management.
- This allows search, like,comment and share functionality.
- being able to use a database well for video data management and storage
- For all users so that they can log in securely and their data privacy is maintained.

4. Scope of the Project

- Has a simple and user-friendly interface for sharing videos online.
- Enables student, content producer and local groups to upload video here (note: live broadcasts will be handled separately)
- supports core features for video-sharing such as uploading, streaming, liking videos while adding comments and subscribing.
- Provides features for video management and secure user authentication.

Focused on the core functionality, avoids some of complexities associated with larger video sharing networks.

Future Scope:

- Incorporation of live streaming capabilities.
- Creation of a version of the mobile application.
- Deployment of video recommendation systems powered by AI.
- Features that allow content providers to make money are added.

5. System Requirements

5.1 Hardware Requirements

- Processor: Intel i3 or above
- RAM: 4 GB or more
- Hard Disk: 50 GB free space (for storing videos and database)
- In order to host and access the platform, an internet connection is necessary.

5.2 Software Requirements

- Operating System: Windows 10 / Linux
- Frontend: HTML, CSS, JavaScript, React
- Backend: Node.js .
- Database: MongoDB / MySQL
- Tools: Visual Studio Code, Git, Postman

6. Proposed Modules

User Module

- User registration, login, and profile management.

Video Upload Module

- Upload, edit, delete, and categorize videos.

Video Playback Module

- Stream videos and view related content.

Interaction Module

- Like, comment, and share videos.

Search Module

- Search videos by title, category, or uploader.

Admin Module

- Manage users, moderate content, and view platform statistics.

7. Working of the System

This is how the system works in a few simple steps:

- User logs in or registers on the platform.
- Add a title, description and category before uploading your video.
- The user is associated to the video that remains on server in db
- Others will be able to explore, search for and view the videos you submit.
- The videos viewed by the viewers can be liked, commented or shared.
- An administrator monitors the platform to ensure content is compliant.

8. Advantages of the Proposed System

- A user-friendly interface suitable for all kinds of users.
- gives up-and-coming content producers a stage on which to present their work.
- can be expanded with new features in the future and modified.
- economical in comparison to big commercial platforms.

9. Conclusion

The objective of the Video Sharing Platform project is to allow users with simple and convenient upload, share, watch videos on the Internet. It offers a simple and easy-to-use interface, allowing users to engage with likes, comments, and subscriptions. This project types practical skillsets such as database administration, program design and well building. Live stream, Mobile app support and AI based video recommendation can be added as next level features which will complement this platform.

10. Gantt chart

Task	W1	W2	W3	W4	W5	W6
Requirement Analysis						
System Design						
Database Design						
Frontend & Backend Development						
Integration & Testing						
Documentation						

11.References

Books:

- React Development: Learning React by A. Banks & E. Porcello (O'Reilly Media).
- Backend Programming: Node.js in Action by M. Cantelon et al. (Manning Publications).
- Database Management: MongoDB: The Definitive Guide by S. Bradshaw et al. (O'Reilly Media).
- Web Fundamentals: Eloquent JavaScript by M. Haverbeke (No Starch Press).

Websites & Documentation:

- React Official Docs: react.dev (For frontend framework implementation).
- Node.js Official Docs: nodejs.org (For backend server configuration).
- MongoDB Manual: mongodb.com/docs (For database architecture).
- MDN Web Docs: developer.mozilla.org (For HTML, CSS, and Web Video APIs).